

Consultant Pharmacists

of
America

Total Parenteral Nutrition(TPN)

Virtual Handbook

Edited

By

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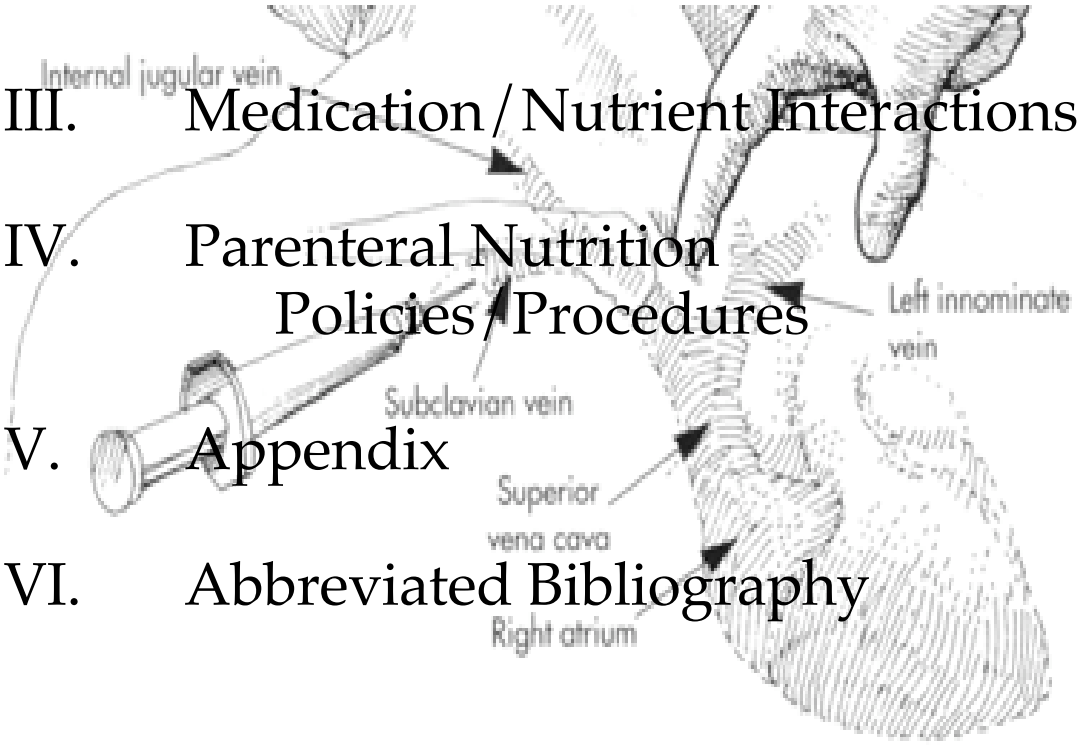
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TPN Therapy
Rounds Presentation
Daily Monitoring Parameters

Nutrition Support Services
_____ Hospital

Bedside Data Collection *- (6 AM until rounding time 9:30 AM)*

Demographic Information Base

1. Name
2. Age/Race/Gender
3. Chief Complaint/Diagnosis
4. Pertinent History (Detailed)/Progress
5. Allergies
6. Day(s) Post admit, Post op, TPN, Enteral
7. Weight (kg) - admit/normal/ recent wt loss/gain
8. Basal Energy Expenditure/Thermodilution/ Metabolic EE (Indirect Calorimetry)

Accumulate morning data for each patient dealing with:

- *day: admit/postop/TPN*
- *HR/MAP (calculate if not available)/BP*
- *IV unfiltered breaks (i.e., any TPN ingredients IV's containing CHO, sodium salts, potassium salts...) administered IV after TPN Hangtime; including AM breaks*
- *Enteral formula(AA and kcal/day)/rate*
- *Diprivan OR propofol rate*
- *CVP/Wedge if available*
- *Insulin drip, SQ, IM*
- *Ventilation (SIMV/IMV/Assist-Control) rate*
- *Net (I-O without insensible losses) /urine output*
- *Other significant output (CT, Duval, stool etc) greater than 500 mL*
- *SIRS/sepsis criteria assessment (if 2 or more criteria - report as hypermetabolic) pCO2, WBC, bands, tachycardia, tachypnea, temp; V.O2/Index*
- *Catheters/tubes; trach/intubation; decubitae*
- *Housestaff Name/cell # (who signed last TPN order)*
- *Reason for parenteral feeding*
- *Progress notes*

Route Assessment

Enteral Monitoring

9. ***Guaiacs/Distention/Transit time (vol diarrhea);***
10. ***Drains; I &O***
11. ***Type enteral tube/distal tip tube placement; formula (rate)***

***Parenteral M*onitoring**

12. ***Access (central/peripheral)***
13. Formula(s) - AA/CHO/Rate; Fat (mL)

MACRONUTRIENTS

Amino acids

14. *Renal Function - BUN/SCr/CrCl*
15. *TP/Alb (LFT)*
16. *Prealbumin, Acute Phase Reactants*
17. *PT/INR*
18. UUN/TUN or Urea-BF
19. Fractional Excretion of Na
20. Unusual protein losses (CVVHD, HD; ascites)
21. Transferrin
22. Ventilatory drive

Carbohydrate

23. *Phosphate/ionized calcium*
24. *Insulin use (IV in units/mL; sliding scale)*
25. *Blood Glucose*
26. *mg dextrose/kg/minute*
27. Glycation Rx/Glycosuria
28. kcal:nitrogen ratio

Fat

29. *Triglyceride*
30. Product content implications (i.e., phosphate, vitamin K, vitamin E, selenium, cholesterol)
31. *Percent fat of total calories*
32. Serum opacity
33. Time of infusion (icosanoid shower)
34. Omega 3 vs Omega 6

MICRONUTRIENTS

Electrolytes

35. *Baumgartner's Rule of Three*
    Step 1. Acid/base Assessment (pH, Base deficit/excess, pCO_2, vent rate);
    Step 2. Abnormals;
    Step 3. Trends
36. Fluids - Serum, Urine (FE of Na, Mg, osmolality, pH etc) and clinical symptoms

Trace Elements

37. *Blood sugar (Cr)*
38. *Dec T3/CPK elevation/Muscle Pain (Se)*
39. *Wound Healing (Zn - avoid during APR).*
40. Ferritin levels (Fe)
41. Uric acid (Mo)
42. *Neutrophil count (Cu)*
43. Wintrobe indices (Fe)
44. Inflammation/infection (Zn, Fe)
45. Other clinical signs/symptoms

Vitamins

46. *Hyperhomocysteinemia (Folic acid)*
47. *L- carnitine (Dialysis pts)*
    Prothrombin Time (Vitamin K)
    Wound Healing (Vit C - no more than 500mg/day); Great AA/CHO load

Nutritional Tx/Medication Implications

48. *Prior to Admission medications*
49. *New medications added each day*
50. *Medication/lab/disease implications*